

When to Order Evacuation under Wildfire Forecast Uncertainty: A Sequential Bayesian Rule against Fixed Triggers in North-East Attica

Chiheb Mejrissi^{a,*}

^a*Department of Computer Science, Example University, 1 Example Street, Example City, Country*

Abstract

Intent: State the decision (order a zone now or wait for a better forecast), the region and hazard, the policies compared, and the B001 outcomes of the headline claims in ARGUMENT.md order; no result before generated/ exists.

Keywords: wildfire evacuation, evacuation triggers, sequential decision under uncertainty, value of information, wildland–urban interface, traffic simulation, responder pre-positioning

1. Introduction

Intent: Motivate the order-now-or-wait decision in a fast wildland–urban interface fire, where waiting sharpens the forecast but shortens the time left to leave, and state the contributions in the headline order of ARGUMENT.md.

2. Related work

Intent: Place the paper against static wildfire triggers, staging optimized for a known fire, and order-now-or-wait rules for other hazards, and say what none of them does; the closest works are baselines and checks [1, 2, 3].

*Corresponding author.

Email address: chiheb.mejrissi@example.org (Chiheb Mejrissi)

3. Region, hazard scenarios and population

Intent: Describe the north-east Attica study region, the two anchor events used for validation only, the generator of training and test fire scenarios, and the population and road network.

3.1. Study region and anchor events

Intent: Why this region, its zones and exits, and the two anchor events with their rapid-mapping products.

3.2. Scenario generator

Intent: How ignitions and fire weather are drawn and how the training and test sets are formed before any policy is run.

3.3. Population, behaviour and traffic

Intent: Residents, households and vehicles per zone, compliance and preparation time, spontaneous departure, and the mesoscopic traffic model.

4. Decision problem

Intent: Define zones, the fire-arrival field, the forecast updated by the observed front (its skill and lead), the per-epoch actions, and the decision cost combining exposure, false-alarm persons and responder time.

4.1. Forecast and its updating

Intent: The member bank, the prior around the true spread parameters, reweighting by agreement with the observed front, and the skill, lead and misspecification settings.

4.2. Actions, outcomes and decision cost

Intent: Irreversible zone orders and responder moves per epoch; exposure, false alarms and unit-hours; the cost ratio fixed before any result.

5. Policies compared

Intent: Define the fixed triggers (buffer, probability threshold, stochastic trigger boundary computed once and updated), the sequential Bayesian rule, simultaneous area-wide evacuation, the learned comparator, the references and the responder arms.

5.1. Fixed triggers

Intent: The buffer trigger, the fixed probability threshold, and the stochastic trigger boundary, computed once and recomputed from the posterior, each tuned on training scenarios.

5.2. Sequential Bayesian rule

Intent: Order a zone when the expected cost of ordering now is below that of waiting one epoch for the updated forecast, congestion included.

5.3. Simultaneous evacuation, learned comparator and references

Intent: The area-wide order family, the learned policy reported without a claim, and the no-order and oracle references.

5.4. Responder pre-positioning

Intent: Static stations from historical ignition density against dynamic staging at the highest posterior risk, at equal unit-hours.

6. Experimental design and evaluation

Intent: State the pre-registered endpoints, their decision rules (supports, weakens, refutes, inconclusive), the multiplicity control, the bootstrap over test scenarios, the sample-size rule, the checks that are not evidence and the anchor validation, all fixed before any result.

6.1. Endpoints and decision rules

Intent: Primary endpoints under family-wise control and the secondary endpoints read outside that family, one per claim.

6.2. Validation and checks

Intent: Anchor validation against rapid-mapping delineations and documented timings, and the checks whose failure is a bug, not evidence.

7. Results

7.1. Sequential rule against the best fixed trigger

Intent: Report the outcome of the sequential rule against the best training-tuned fixed trigger on held-out scenarios, with the misspecification and cost-ratio arms beside it.

7.2. Value-of-information ladder

Intent: Decompose the gain into updating the forecast, the change of rule form (reported only) and planning, and the perfect-information gap left.

7.3. Lead time against forecast skill

Intent: Report which lever lowers the cost more in fast and in slow scenarios, and how the advantage of lead over skill varies with the evacuation safety margin.

7.4. Forecast error and order breadth

Intent: Report whether the advantage of staged over simultaneous evacuation, at equal false alarms, changes when the forecast error is halved.

7.5. Staged against simultaneous evacuation (replication)

Intent: Report the staged-versus-simultaneous comparison at equal false alarms as a replication, with the descriptive non-monotonicity check.

7.6. Responder pre-positioning (secondary)

Intent: Report dynamic against static pre-positioning at equal unit-hours, with its dependence on the assumed responder effect stated.

7.7. Anchor events (descriptive)

Intent: Describe the anchor audits: consistency with published evacuation-safety findings for the first anchor, the order-time audit for the second, and the cross-model clearance check.

8. Discussion

Intent: Interpret the outcomes for evacuation practice: what forecast updating and planning are worth, when lead time beats forecast skill, and why broad early orders can congest the exits the threatened zones need.

9. Limitations

Intent: State that exposure is a proxy for harm, that results do not transfer to other regions or hazards without a new study, that historical decisions are not judged, and that the responder effect is an assumption.

10. Conclusion

Intent: Summarize the outcomes of the headline claims in ARGUMENT.md order and what a fire service can take from them.

Data and code availability

The simulation code, the scenario and experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). The input data are public: road networks from OpenStreetMap, rapid-mapping products from the Copernicus Emergency Management Service, fire records from the European Forest Fire Information System, reanalysis weather from the Copernicus Climate Data Store and population grids from the Global Human Settlement Layer; access dates and snapshots are listed in the deposit.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the simulation and analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

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